

Use Case LM–1.1: Initialize Rocket**Scope:** Launch Mission**Primary Actor:** Flight Director**Secondary Actor:** None**Related Use Cases:** *Use Case 1 Initialize Launch,*
Use Case 1.1.1 Initialize Stage,
*Use Case 1.1.2 Initialize Capsule***Stake Holders & Interests:**

- Flight Director: Wants to set up a Launch via initializing all the Launch Parameters, Systems, Sub-Systems including:
 - Rocket
- Mission Planner: Wants the Mission Initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The System is running and ready to initialize and monitor the appropriate Systems and Sub-Systems**Post-Conditions:** The Rocket and all the components is initialized with the appropriate parameters**Flow of Events:**

Flight Director	System
1. Enters the Mission Parameters File (See <i>Use Case 1 Initialize Launch</i>)	
	2. Initialize each Stage (See <i>Use Case 1.1.1 Initialize Stage</i>)
	3. Initialize the Payload/Capsule (See <i>Use Case 1.1.2 Initialize Capsule</i>)
	4. Finds the Rocket Parameters file name and location
	5. Goes to the Rocket Parameters file location
	6. Opens the Rocket Parameters file
	7. Reads the Rocket Parameters file
	8. Initializes the Rocket Data with the appropriate Rocket Data read in from the Rocket Parameters file
	9. Queries or receives Initialization Status from each Stage (See <i>Use Case 1.1.1 Initialize Stage</i>)
	10. Queries or receives Initialization Status from the Payload/Capsule (See <i>Use Case 1.1.2 Initialize Capsule</i>)
11. Receives Initialization Data	

Extensions (or Alternative Flows):

- 2a If the System receives an Initialization Error from any of the Stages, then the System returns the Error to the Flight Director (See *Use Case 1.1.1 Initialize Stage*). The Flight Director determines the next course of action.
- 3a If the System receives an Initialization Error from the Payload/Capsule, then the System returns the Error to the Flight Director (See *Use Case 1.1.2 Initialize Capsule*). The Flight Director determines the next course of action.
- 4a,5a If the System cannot find the location where the Rocket Parameters File is located, then the System returns an Error to the Flight Director indicating a non-legitimate file path. The Flight Director determines the next course of action.
- 6a If the System cannot open the Rocket Parameters file, then the System returns an Error to the Flight Director indicating unable to open the Rocket Parameters file. The Flight Director determines the next course of action.
- 7a If the System cannot read the Rocket Parameters file, then the System returns an Error to the Flight Director indicating unable to read the Rocket Parameters file. The Flight Director determines the next course of action.
- 9a If the System receives an Error from any of the Stages, then the System returns the Error to the Flight Director (See *Use Case 1.1.1 Initialize Stage*). The Flight Director determines the next course of action.
- 10a If the System receives an Error from the Capsule, then the System returns the Error to the Flight Director (See *Use Case 1.1.2 Initialize Capsule*). The Flight Director determines the next course of action.

Notes/Constraints:

- *a This Use Case documents the Rocket Initialization, not the Rocket Monitoring (See *Use Case 4.1 Monitor Initialization*)
- *b The Next Course of action for the Flight Director is aborting the mission or choosing another Mission Parameters File.
- 11a The System indirectly alerts the Stake Holders via an upward cascade of alerts/queries (See *Use Case 1.1.1 Initialize Stage*, *Use Case 1.1.2 Initialize Capsule*)

Special Requirements:

- None

Technology & Variations List:

- There could be many types of Mission Parameters Files in different formats for the System Composition. The System needs to account for all the different types of files and read them accordingly.
- Regardless of the type of file, the System Data should follow a standard

Frequency of Occurrence:

- Every time a Mission is planned or simulated

Open Issues:

- None